

Hany Hamed Aly

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Interested in Robot Learning, Multi-Robot Systems & Sim2Real

EDUCATION

Innopolis University

*Bachelor of Computer Science - Robotics track with **Full scholarship***

CGPA: 4.605/5 = 3.684/4 ([Transcript](#))

Innopolis, Russia

Aug 2018 – June 2022

RESEARCH EXPERIENCE

Bachelor Thesis | *In progress*

May 2021 – June 2022

- Topic: **Learning behavioural strategies for drones in Pursuit and Evasion Games using Reinforcement Learning**
- Keywords: Reinforcement Learning, Evolutionary algorithms, Pursuit & Evasion Games, Competitive coevolution, Self-play
- Supervised by: Prof. [Stefano Nolfi](#) & Prof. [Alexandr Klimchik](#)

Undergraduate Research Assistant

Aug 2019 – April 2021

Center for Technologies in Robotics and Mechatronics Components, Innopolis University

Innopolis, Russia

- Developed a gym environment for a Tensegrity hopper
- Conducted experiments using [ARS](#) algorithm to learn a stabilizing control policy for the tensegrity hopper
- Implemented a contactless differentiable physics simulator for tensegrity robots using [Taichi](#)
- Developed experiments to make different tensegrity robots learn reaching specific configurations in simulator and transfer the control policy to the real robot

Summer Research Intern

June 2019 – July 2019 – [[Doc](#)]

Center for Technologies in Robotics and Mechatronics Components, Innopolis University

Innopolis, Russia

- Explored the simulation of different robots in NTRTsim (NASA simulator for Tensegrity Structured robots)
- Implemented an integration between NTRTsim and python to be used with learning algorithms
- Explored the possibility to teach a tensegrity robot to perform rolling locomotion using Reinforcement Learning
- Developed a CAD model for a simple tensegrity prototype

PUBLICATIONS & IN PROGRESS MANUSCRIPTS

- G. Kulathunga, H. Hamed, D. Devitt and A. Klimchik, “Optimization-based Trajectory Tracking Approach for Multi-rotor Ariel Vehicles in Unknown Environments” [Manuscript submitted for publication]
- V. Kurenkov, H. Hamed, and S. Savin, “Learning stabilizing control policies for a tensegrity hopper with augmented random search,” in *2020 International Conference on Industrial Engineering, Applications and Manufacturing (ICIEAM)*, pp. 1–5, IEEE, 2020 ([code](#), [paper](#))
- L. Vorochaeva, S. Savin, H. Hamed, and A. M. Leon, “Analysis of algorithms for controlling the length of crawling robot modules,” in *2020 4th Scientific School on Dynamics of Complex Networks and their Application in Intellectual Robotics (DCNAIR)*, pp. 257–260, IEEE, 2020 ([paper](#))
- L. Vorochaeva, S. Savin, and H. Hamed, “Lateral gait analysis of a crawling robot by means of controlling the lengths of links and friction in the supports,” in *2020 International Conference Nonlinearity, Information and Robotics (NIR)*, pp. 1–6, IEEE, 2020 ([paper](#))
- H. Hamed, V. Kurenkov and S. Savin, “Differentiable Tensegrity Simulator with Sim2Real Experiments” [Manuscripts in progress]
- H. Hamed, S. Savin and L. Vorochaeva “A Brief Survey on Locomotion Control Methods For Crawling Robots” [Manuscripts in preparation]

WORKING EXPERIENCE

Laboratory Assistant

June 2021 – Present

Unmanned Technology Laboratory

Innopolis, Russia

- Integrating and developing ROS packages for different modules for drones (Gimbal control, RTK GPS, ...etc)
- Conducting experiments with different drones
- Developing a handheld lidar for indoor mapping for an industrial partner
- Developing a robot for indoor navigation for an industrial partner

Technical Intern

July 2019 – Aug 2019 - [\[Doc\]](#)

Copter Express

Moscow, Russia

- Had a training about main components and construction of drones
- Had a training on COEX's autonomous drones (ROS, PX4 and indoor navigation by Aruco markers)
- Developed a human pose estimation with a webcam using Tensorflow.js to control a drone
- Integrated RTAP-Map SLAM algorithm with COEX's drone (Clever 3) using Intel Real-Sense T265 and D435 cameras

PROJECTS

Human Pose Estimation Drone Control | *Python, TF.js, ROS*

July 2019

- Explored and tested different human pose estimation modules
- Integrated a drone controller with human pose estimation module built with TensorFlow.js

Automated 3D Scanner (Team Project) | *Python, ROS, Gazebo*

Jan. 2019 – May 2019

- Reviewed different existing solutions for 3D scanners
- Collaborated on developing the base idea for the 3D scanner
- Collaborated on developing a simulation for the 3D scanner
- Integrated a keyboard teleop with the simulator using ROS to control the motion of the 3D scanner
- Explored the usage of Open3d for colored point cloud registration

SELECTED AWARDS & CERTIFICATES & SUMMER SCHOOLS

European Agent Systems Summer School (EASSS 2021) | Student

July 2021

3rd place DOTS competition (Simple Solution team) | [Competition report](#) | [Certificate](#)

June 2021

Outstanding Contribution to Science at Innopolis University during 2020/2021 | [Certificate](#)

June 2021

Outstanding Achievements at Innopolis University for Fall 2020 | [Diploma](#)

January 2021

Winter School on Machine Learning in Robotics | [Certificate of Participation](#)

Dec 2020

[Coursera] Modern Robotics, Course 1: Foundations of Robot Motion | [License](#)

Jan 2019

[Coursera] Modern Robotics, Course 2: Robot Kinematics | [License](#)

Jan 2019

EXTRACURRICULAR ACTIVITIES

Leader for RoboCup Small Size League (SSL) Innopolis team

February 2021 – Present

- Constructed a team for participation
- Reviewed the existing solutions from different teams
- Constructed the bill of materials for team's equipment
- Supervised, guided and helped other members in the sub-teams (Mechanical, electrical and control)
- Prepared plans and tasks for the team's members
- Developed a basic controller for the robots in the simulation

Judge at WRO International Finals [Advanced Robotics Challenge - ARC]

2019

- Developed the game rules for the international competition
- Judged the Russian qualifications as the Head judge
- Judged the International competition
- Collaborated in the development of a playground generator for the competition using C++ and Javascript

Participant in RoboCup Junior Egypt Open Weight Soccer Category

2018

- Developed a colored ball detection software using OpenCV
- Developed PCB boards for the robot
- Developed the game controller for the robots
- Won 2nd place